

# Literature overview on Graph Neural Networks for Time Series Forecasting

Guillermo Martin

April 2025

'Graph Deep Learning for Time Series Forecasting' by Andrea Cini  
et al.

<https://arxiv.org/pdf/2310.15978>

# Setting

- ▶  $N$  correlated time series on graph nodes.
- ▶  $X_t \in \mathbb{R}^{N \times d_x}$ , exogenous  $U_t$ , static features  $V$ .
- ▶ Adjacency matrix  $A_t$  or edge set  $E_t$ .

$$x_t^i \sim p_i(x_t^i \mid X_{<t}, U_{\leq t}, V)$$

- ▶ Predict  $X_{t:t+H}$  from past window  $W$ .

$$p_\theta(x_{t+h}^i \mid G_{t-W:t}, U_{t:t+h+1})$$

$$\hat{X}_{t:t+H} = F(G_{t-W:t}, U_{t:t+h+1}; \theta)$$

$$\theta^* = \arg \min_{\theta} \frac{1}{T} \sum_t \ell(\hat{X}_{t:t+H}, X_{t:t+H})$$

# Spatiotemporal GNNs

Message Passing:

$$h_i^{l+1} = \text{Upd}^l \left( h_i^l, \text{Aggr}_{j \in \mathcal{N}(i)} \{ \text{Msg}^l(h_i^l, h_j^l, e_{ji}) \} \right)$$

Spatiotemporal Extension:

$$h_{i,t}^{l+1} = \text{Upd}^l \left( h_{i,\leq t}^l, \text{Aggr}_{j \in \mathcal{N}(i)} \{ \text{Msg}^l(h_{i,\leq t}^l, h_{j,\leq t}^l, e_{ji,\leq t}) \} \right)$$

# Experimental Findings

- ▶ STGNNs outperform RNNs on GPVAR.
- ▶ GPVAR model:

$$H_t = \sum_{l=1}^L \sum_{q=1}^Q \Theta_{q,l} A^{l-1} X_{t-q}$$

$$X_{t+1} = a \odot \tanh(H_t) + b \odot \tanh(X_{t-1}) + \eta_t$$

# Open Challenges

- ▶ Scalability
- ▶ Latent graph inference.
- ▶ Design choices.

# Torch Spatiotemporal

- ▶ Library for neural spatiotemporal data processing focusing on GNNs
- ▶ Jupyter Notebook on GPVAR

'Deep Learning for Time Series Forecasting: Tutorial and Literature Survey' by Konstantinos Benidis et al.  
<https://arxiv.org/pdf/2004.10240>



| Study                | Structure       | Forecast | Horizon | Loss                            | Metrics                 | Data Types                                             | Comments                                                                                             |
|----------------------|-----------------|----------|---------|---------------------------------|-------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| DeepAR [156]         | RNN             | Prob     | 1       | NLL                             | Coverage, QL, ND, NRMSE | demand, traffic, electricity                           | Learns parametric distributions                                                                      |
| Toubeau et al. [179] | RNN/CNN         | Prob     | 1       | NLL/QL                          | RMSE, price             | electricity                                            | Nonparametric copula to capture multivariate dependence                                              |
| Salinas et al. [155] | RNN             | Prob     | 1       | NLL                             | QL, MSE                 | electricity, traffic, exchange rate, solar, taxi, wiki | Learns multivariate model via low-rank Gaussian copula processes                                     |
| ARMDN [136]          | RNN             | Prob     | 1       | NLL                             | wMAPE                   | demand                                                 | Like [156], but using mixture of Gaussian's and domain specific feature processing                   |
| QARNN [196]          | MLP             | Prob     | 1       | QL                              | VaR, ES                 | finance                                                | Conditional quantile function estimation                                                             |
| SQF-RNN [64]         | RNN             | Prob     | 1       | CRPS                            | QL, MSIS, NRMSE, OWA    | demand, traffic, count data, finance, M4               | Models non-parametric distributions with splines                                                     |
| LSTNet [107]         | CNN + RNN + MLP | Point    | 1       | $\ell_1$                        | RRSE, CORR              | traffic, solar, electricity, exchange rate             | Extracts short and long temporal patterns with a CNN and RNN, respectively                           |
| Zhu and Laptev [205] | RNN + MLP       | Prob     | 1       | -                               | sMAPE, calibration      | daily trips                                            | Fits an encoder (RNN) that constructs an embedding state, which is fed to a prediction network (MLP) |
| Laptev et al. [111]  | RNN             | Prob     | 1       | MSE                             | sMAPE                   | traffic, M3                                            | LSTM as feature extractor                                                                            |
| Qiu et al. [144]     | MLP + SVR       | Point    | 1       | $\ell_2$ for MLP, SVR objective | RMSE, MAPE              | energy, housing                                        | Ensemble of DBNs where their output is fed to an SVR                                                 |
| A-LSTM [82]          | RNN + MLP       | Point    | 1       | $\ell_2, \ell_2$ regularizer    | RMSE                    | electricity consumption                                | Combination of LSTM with autoencoders                                                                |
| Borovykh et al. [24] | CNN             | Point    | 1       | $\ell_1, \ell_2$ regularizer    | RMSE, MASE, HITS        | index forecasting, exchange rate                       | WaveNet [183] based model adjusted for time series forecasting                                       |

'Graph Neural Networks for temporal graphs: State of the art, open challenges, and opportunities' by Antonio Longa, Pietro Liò et al.  
<https://arxiv.org/pdf/2302.01018>

# State of the Art

- ▶ They consider different tasks such as Temporal Node/Edge/Graph Classification, Temporal Link Prediction, Temporal Node Clustering, and Regression.

# Temporal GNN Architectures Taxonomy

- ▶ Snapshot-based Models
  - ▶ Model Evolution: Evolving GNN parameters across time (e.g., EvolveGCN)
  - ▶ Embedding Evolution: Evolving node embeddings (e.g., DySAT, VGRNN, ROLAND)
- ▶ Event-based Models:
  - ▶ Temporal Embedding Methods: Use time encodings with recurrent/self-attention updates (e.g., TGAT, NAT)
  - ▶ Temporal Neighborhood Methods: Use mailboxes to store event information and update nodes (e.g., TGN, DGNN, TGL)

# Open Challenges

- ▶ Lack of standardized benchmarks (like Open Graph Benchmark for static GNNs)
- ▶ Need for deeper understanding of expressiveness (e.g., temporal Weisfeiler–Lehman tests)
- ▶ Handling learnability issues (over-smoothing, over-squashing)
- ▶ Extensions to regression tasks.

'Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks' by Michael M. Bronstein, Christopher Morris et al.  
<https://arxiv.org/pdf/2502.14546>

- ▶ It is argued that graph learning risks becoming less relevant because current benchmarking practices are fundamentally flawed. Benchmarks often focus on narrow, unrealistic tasks, and lack standardization.
- ▶ They remark that no true foundation model exists for graph learning. Domain-specific graphbased models have appeared recently but performance often shows only marginal gains over standard supervised GNNs.
- ▶ They propose shifting from one model for one dataset to one model for all datasets to provide a comprehensive view of model performance across diverse graphs.

## Over-smoothing and Over-squashing



# Over-smoothing

- ▶ Graph convolutions usually use smoothing functions on each layer. Therefore, when we apply several graph convolution layers, the performance can suffer from over-smoothing, where node embeddings from different clusters become mixed up.
- ▶ Over-smoothing lacks a formal definition in the literature.

# Over-squashing

- ▶ Over-squashing: large amount of information from a big neighborhood needs to be compressed into a fixed-size node representation. As a result, important information from distant or even nearby nodes can get "squashed" or lost during message passing.
- ▶ Exponential decay of the node dependence on input features at distance.

# Literature

- ▶ On the Trade-off between Over-smoothing and Over-squashing in Deep Graph Neural Networks.  
<https://arxiv.org/pdf/2212.02374>
- ▶ A Survey on Oversmoothing in Graph Neural Networks.  
<https://arxiv.org/pdf/2303.10993>
- ▶ Unifying over-smoothing and over-squashing in graph neural networks: A physics informed approach and beyond  
<https://arxiv.org/pdf/2309.02769>

# Temporal Graph Learning



Learning production functions for supply chains with graph neural networks.

<https://arxiv.org/abs/2407.18772>



Graph Deep Learning for Time Series Forecasting (Andrea Cini).

<https://arxiv.org/pdf/2310.15978>



Graph Neural Networks for temporal graphs: State of the art, open challenges, and opportunities.

<https://arxiv.org/pdf/2302.01018>



Deep Learning for Time Series Forecasting: Tutorial and Literature Survey.

<https://arxiv.org/pdf/2004.10240>



Temporal graph kernels for classifying dissemination processes.

<https://arxiv.org/abs/1911.05496>



Temporal Graph Reading.

<https://shenyanghuang.github.io/rg.html>

# Graph Neural Networks



How Powerful are Graph Neural Networks.

<https://arxiv.org/pdf/1810.00826>



On the Trade-off between Over-smoothing and Over-squashing in Deep Graph Neural Networks.

<https://arxiv.org/pdf/2212.02374>



GNNExplainer: Generating Explanations for Graph Neural Networks.

<https://arxiv.org/abs/1903.03894>



On the expressive power of graph neural networks: beyond the Weisfeiler-Leman test.

[https://usiena-air.unisi.it/bitstream/11365/1265054/3/phd\\_unisi\\_103967.pdf](https://usiena-air.unisi.it/bitstream/11365/1265054/3/phd_unisi_103967.pdf)



On the Expressive Power of Geometric Graph Neural Networks.

<https://arxiv.org/pdf/2301.09308>



Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (Michael Bronstein).

<https://arxiv.org/abs/2104.13478>

# Networks



Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks.

<https://arxiv.org/pdf/2502.14546>



Motifs in Temporal Networks.

<https://dl.acm.org/doi/pdf/10.1145/3018661.3018731>



Echoes of the hidden: Uncovering coordination beyond network structure.

<https://arxiv.org/pdf/2504.02757>

# Generalization of Graph Neural Networks



Optimisation & Generalisation in Networks of Neurons.

<https://arxiv.org/abs/2210.10101>



Survey on Generalization Theory for Graph Neural Networks  
(Christopher Morris).

<https://arxiv.org/pdf/2503.15650>



Generalization of Graph Neural Networks through the Lens of  
Homomorphism.

<https://arxiv.org/pdf/2403.06079>



Generalization of Graph Neural Networks is Robust to Model  
Mismatch.

<https://arxiv.org/html/2408.13878v2>



Out-Of-Distribution Generalization on Graphs: A Survey.

<https://arxiv.org/pdf/2202.07987>



Towards Understanding Generalization of Graph Neural Networks.

<https://proceedings.mlr.press/v202/tang23f/tang23f.pdf>